



FSD MINI PROJECT

TEAM MEMBERS	ROLL NO
MANAV SHETTY	10279
YASH SOLANKI	10282
SHONIT SEQUEIRA	10277

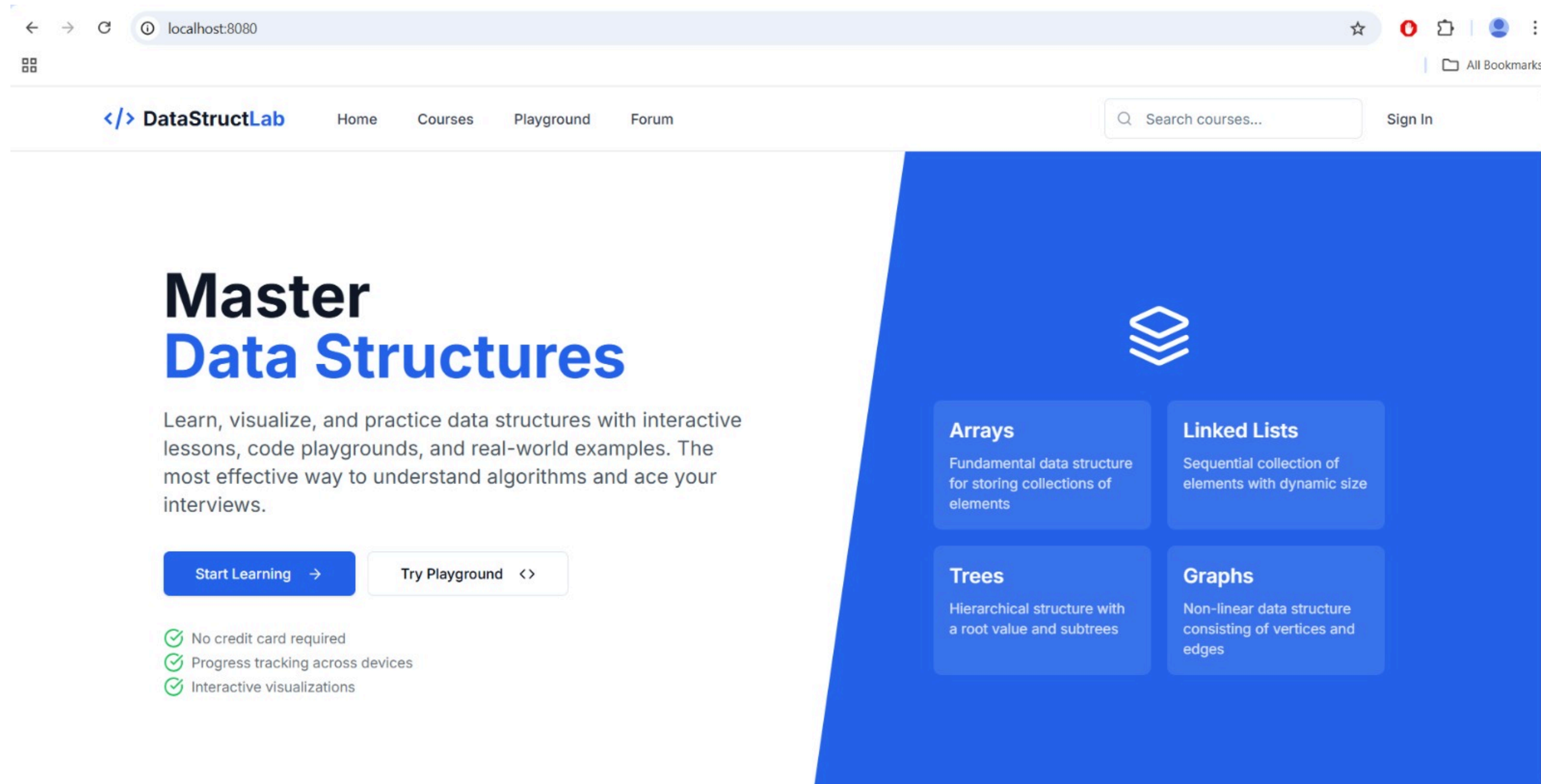
BRANCH - COMPS B

BATCH - D

TOPIC - ITL(Innovative Teaching & Learning)

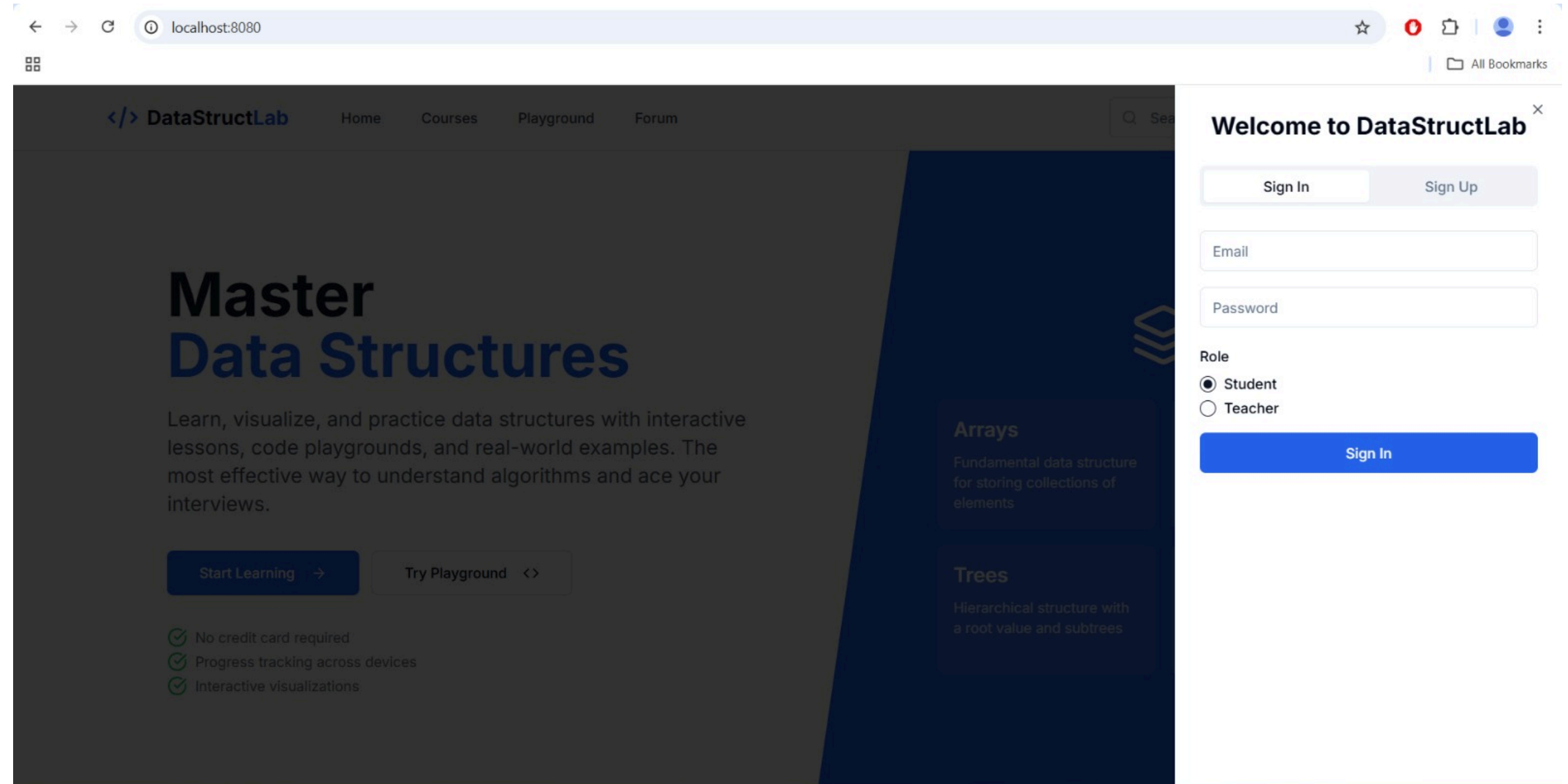


LANDING PAGE



LOGIN POP-UP

Student Login along with Teacher Login



The screenshot shows a web browser at localhost:8080 displaying the DataStructLab website. The website has a dark theme with a navigation bar containing links for Home, Courses, Playground, and Forum. The main content area features the title "Master Data Structures" and a description: "Learn, visualize, and practice data structures with interactive lessons, code playgrounds, and real-world examples. The most effective way to understand algorithms and ace your interviews." Below this, there are two buttons: "Start Learning" and "Try Playground". A list of features is shown at the bottom: "No credit card required", "Progress tracking across devices", and "Interactive visualizations". On the right side, a "Welcome to DataStructLab" pop-up is visible, containing fields for Email and Password, a Role selection (Student or Teacher), and a Sign In button.

localhost:8080

</> DataStructLab Home Courses Playground Forum

Master Data Structures

Learn, visualize, and practice data structures with interactive lessons, code playgrounds, and real-world examples. The most effective way to understand algorithms and ace your interviews.

[Start Learning](#) [Try Playground](#)

- ✓ No credit card required
- ✓ Progress tracking across devices
- ✓ Interactive visualizations

Arrays
Fundamental data structure for storing collections of elements

Trees
Hierarchical structure with a root value and subtrees

Welcome to DataStructLab

[Sign In](#) [Sign Up](#)

Email

Password

Role

☒ Student ☐ Teacher

[Sign In](#)

COURSES PAGE

localhost:8080/courses

All Bookmarks

</> DataStructLab

Home

Courses

Playground

Forum

Q Search courses...

Welcome, manavshetty

Logout

All Courses

Explore our comprehensive collection of data structure courses, from fundamentals to advanced concepts.

Start Your Learning Journey

Explore our comprehensive data structure courses designed for all skill levels.

All

Beginner

Intermediate

Advanced

[]

Arrays and Strings

Beginner

Learn about arrays, dynamic arrays, and string manipulation techniques with practical examples.

🔗

Linked Lists

Beginner

Understand singly and doubly linked lists, their operations, and common patterns and techniques.

📦

Stacks and Queues

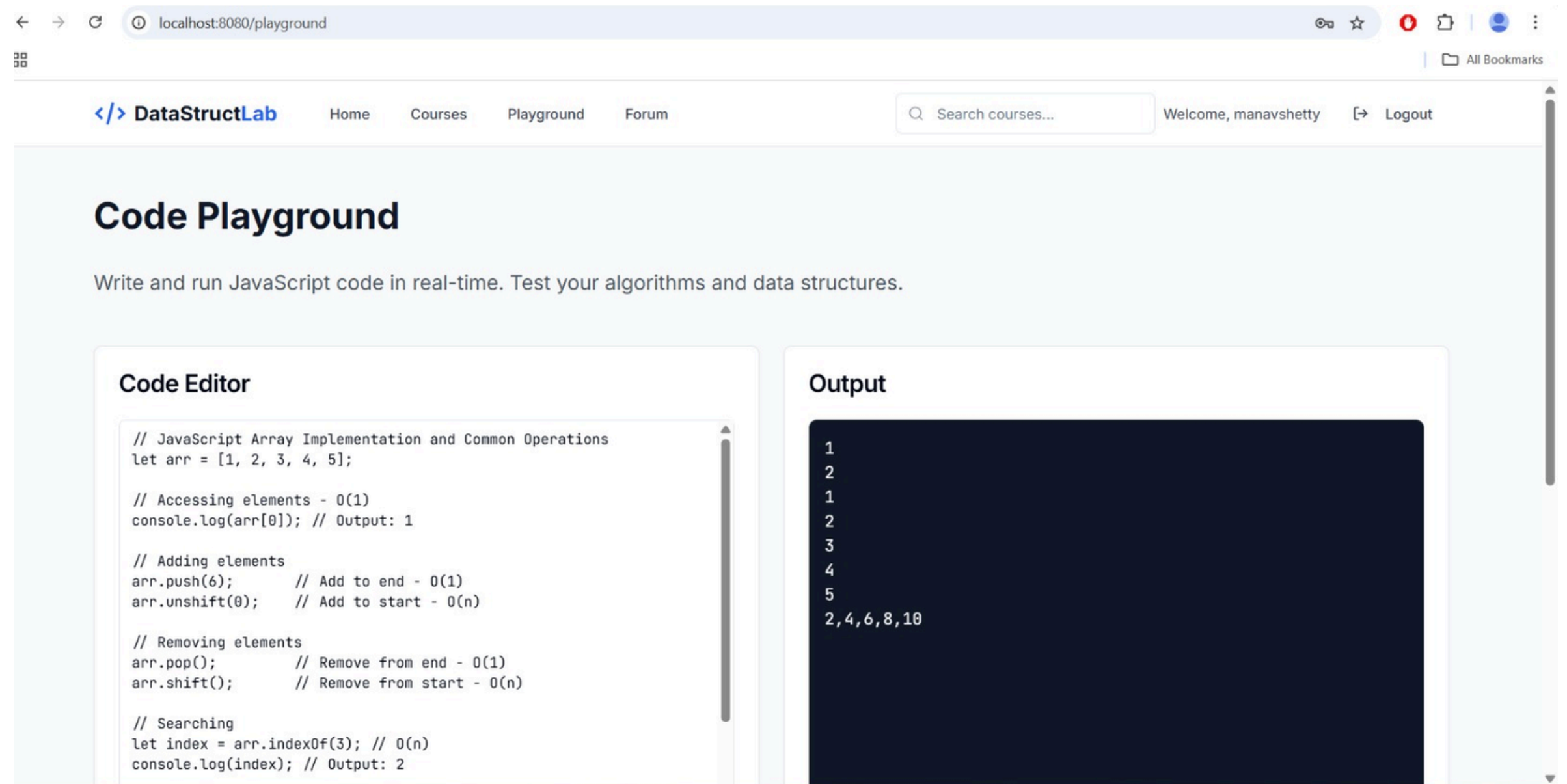
Beginner

Learn about stack and queue data structures, their implementations, and practical applications.



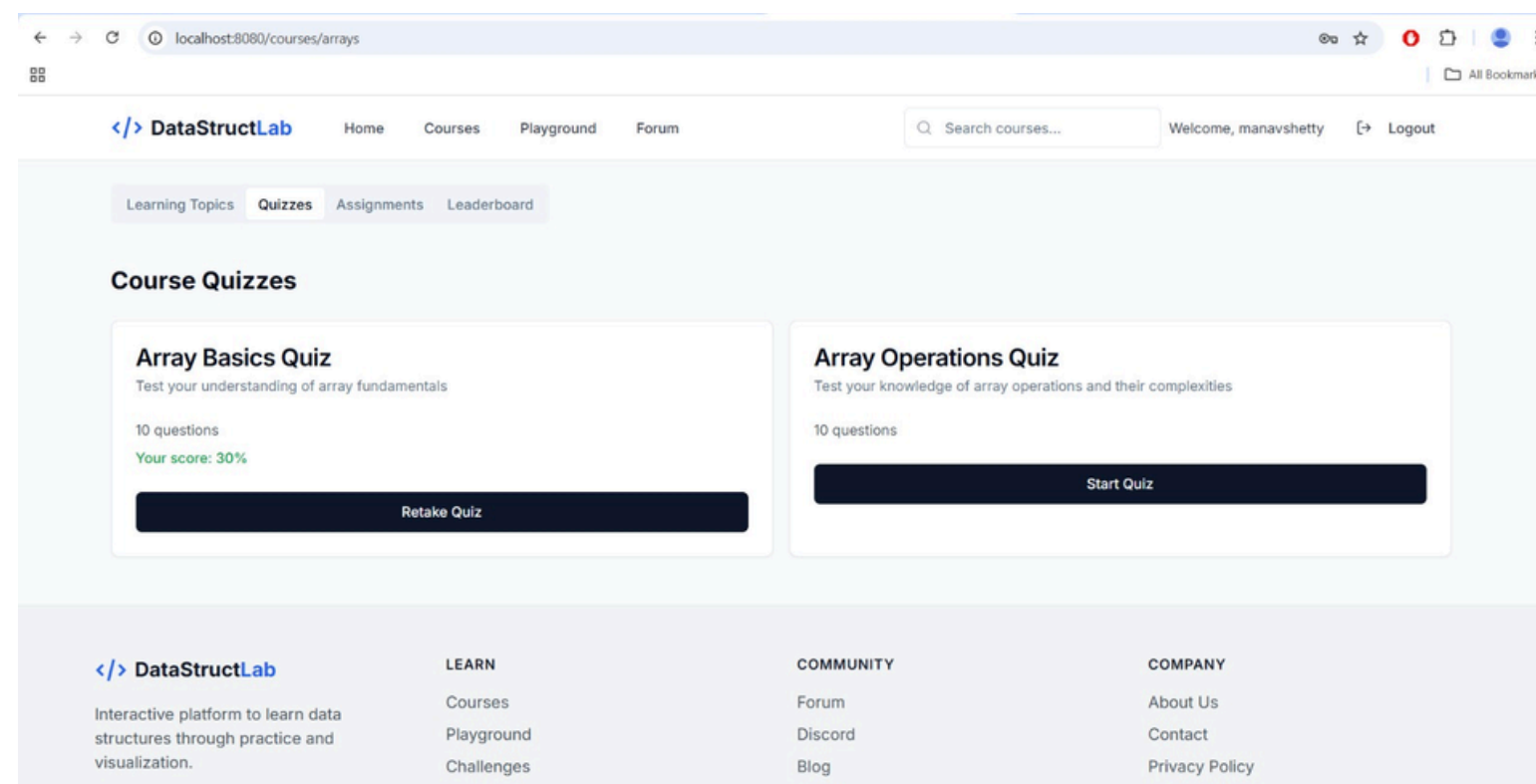
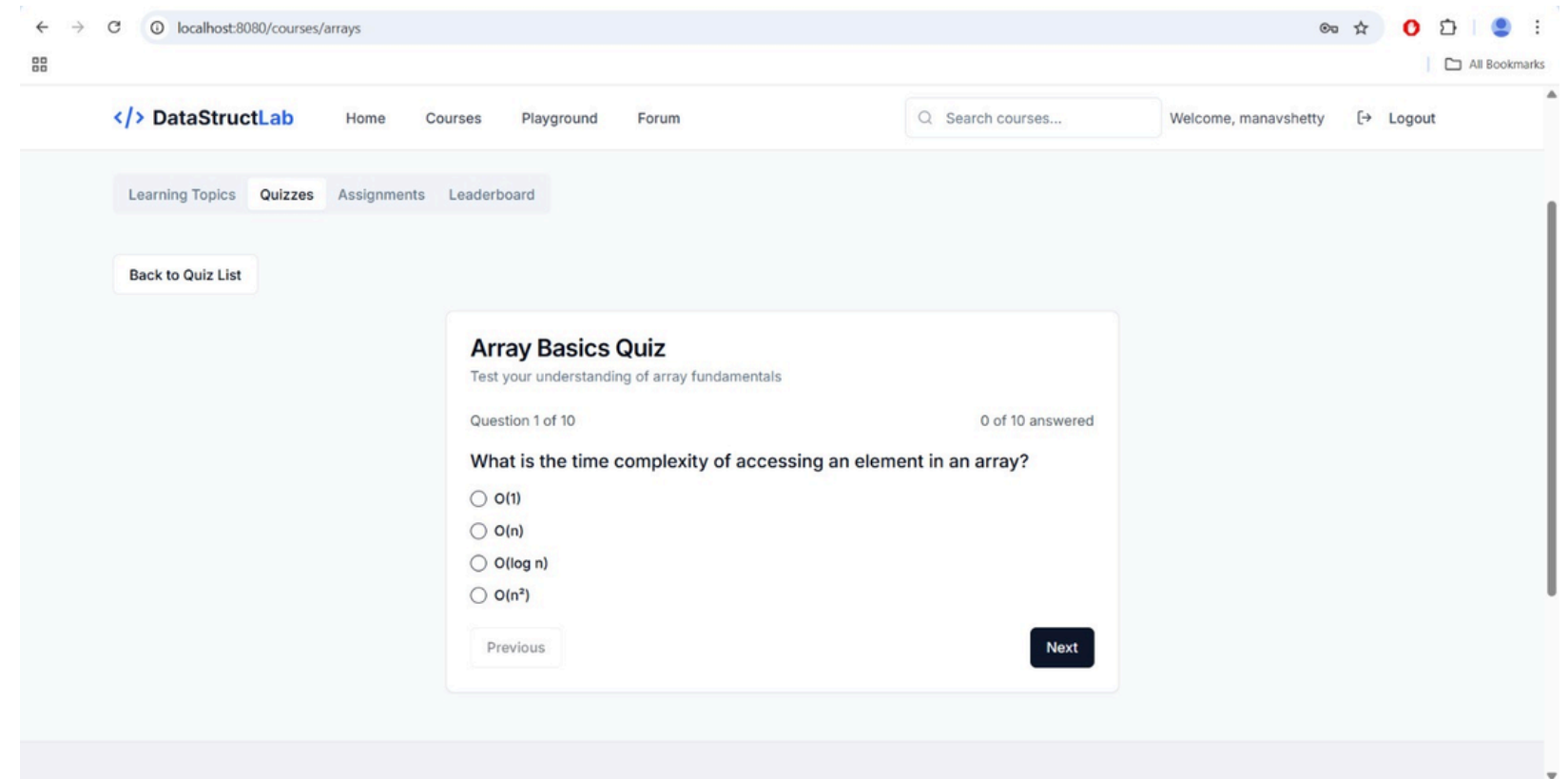
CODE PLAYGROUND

A Code
Editor for
Students to
practice



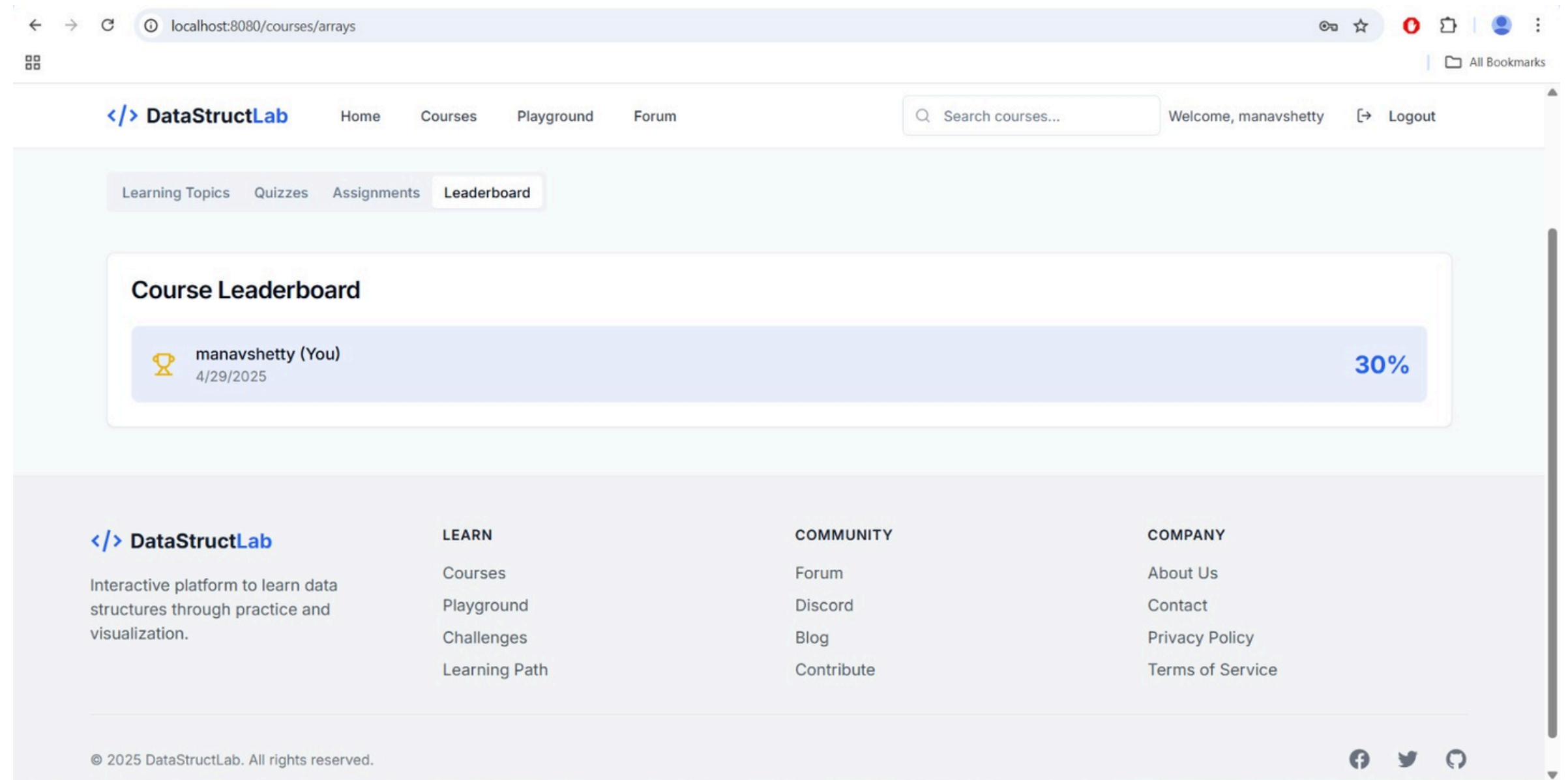
STUDENT LOGIN : QUIZZES

Basic quiz
on the
subjects,
with score
provided
after the test



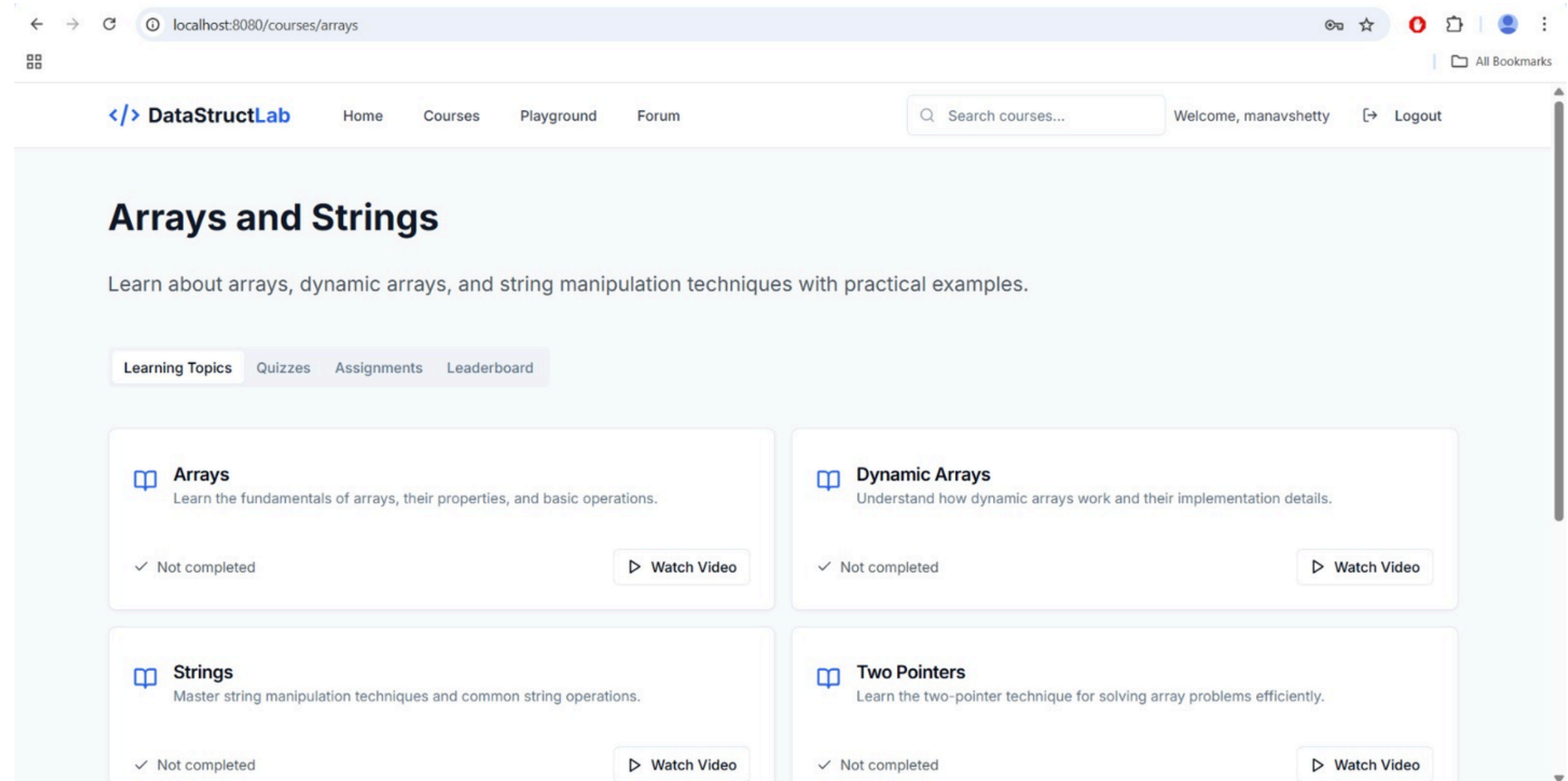
STUDENT LOGIN : LEADERBOARD

Leaderboard
showcasing
the score
along with
name of the
students.



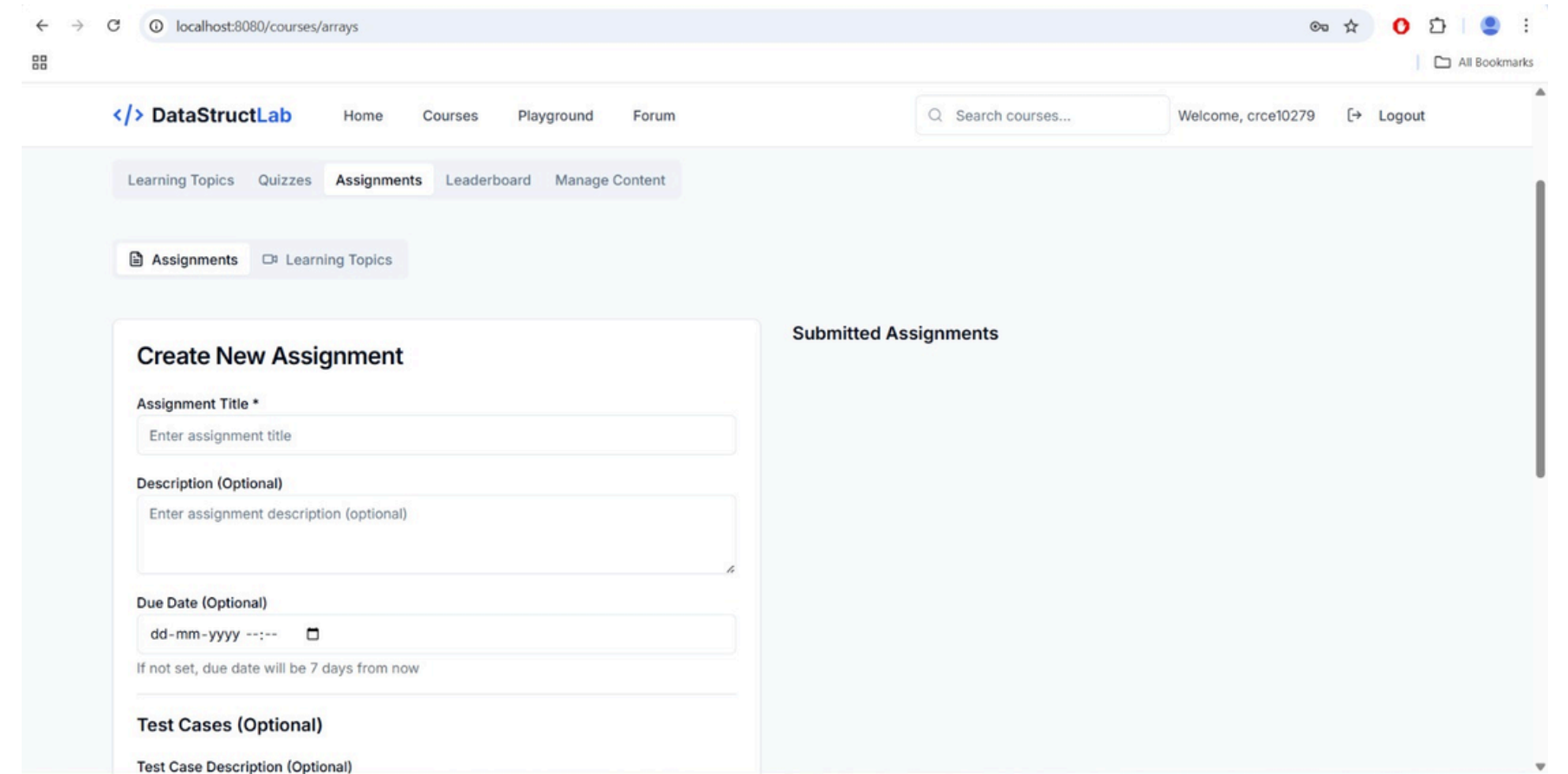
STUDENT LOGIN : LEARNING TOPICS

Videos
uploaded by
the teacher
are visible
here.

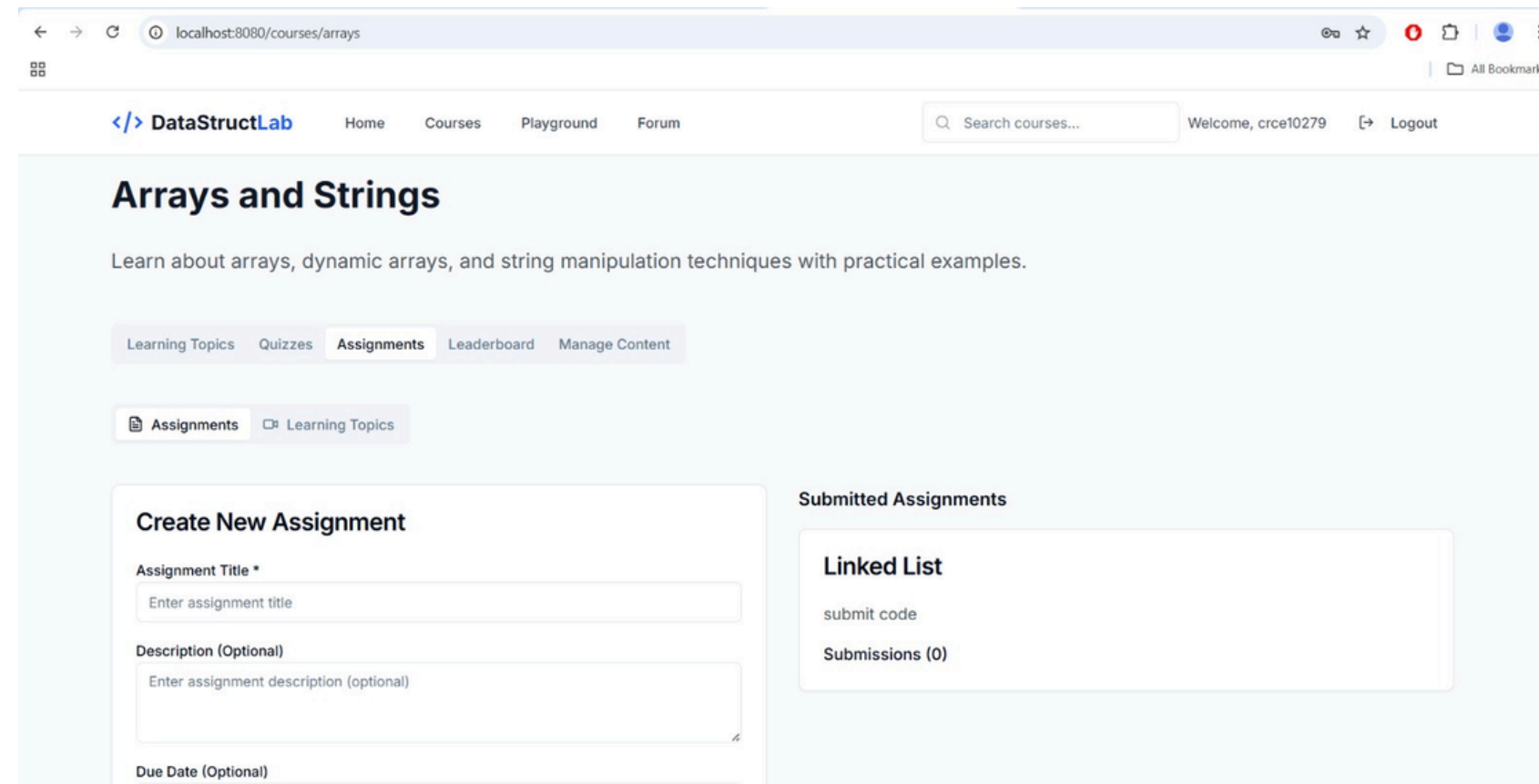


TEACHER LOGIN : ASSIGNMENTS

Allows
teacher to
create new
assignment
and upload
new videos



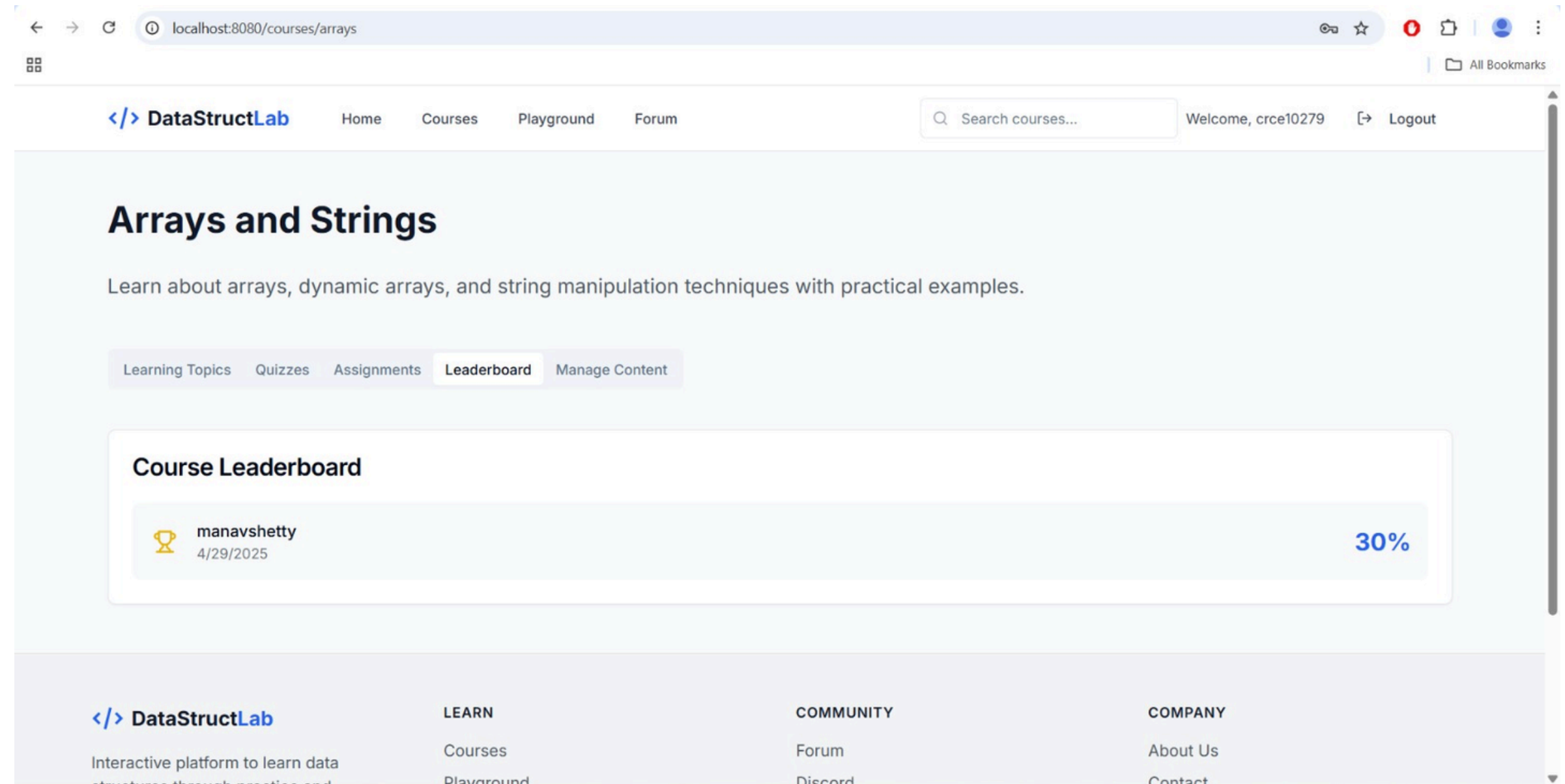
This screenshot shows the 'Create New Assignment' form in the DataStructLab application. The browser address bar indicates the URL is localhost:8080/courses/arrays. The page has a navigation bar with links for Home, Courses, Playground, and Forum, along with a search bar and a user profile section showing 'Welcome, crce10279' and a 'Logout' button. Below the navigation bar, there are tabs for 'Learning Topics', 'Quizzes', 'Assignments' (which is active), 'Leaderboard', and 'Manage Content'. On the left side of the main content area, there are sub-tabs for 'Assignments' and 'Learning Topics'. The 'Create New Assignment' form includes the following fields: 'Assignment Title *' (required), 'Description (Optional)', 'Due Date (Optional)' (with a date picker and a note that it defaults to 7 days from now), and 'Test Cases (Optional)' (with a 'Test Case Description (Optional)' sub-field). On the right side, there is a section titled 'Submitted Assignments'.



This screenshot shows the 'Arrays and Strings' course page in the DataStructLab application. The browser address bar indicates the URL is localhost:8080/courses/arrays. The page has a navigation bar with links for Home, Courses, Playground, and Forum, along with a search bar and a user profile section showing 'Welcome, crce10279' and a 'Logout' button. Below the navigation bar, there are tabs for 'Learning Topics', 'Quizzes', 'Assignments' (which is active), 'Leaderboard', and 'Manage Content'. On the left side of the main content area, there are sub-tabs for 'Assignments' and 'Learning Topics'. The 'Create New Assignment' form is visible on the left, with fields for 'Assignment Title *', 'Description (Optional)', 'Due Date (Optional)', and 'Test Cases (Optional)'. On the right side, there is a section titled 'Submitted Assignments' which contains a 'Linked List' assignment with a 'submit code' button and a 'Submissions (0)' indicator.

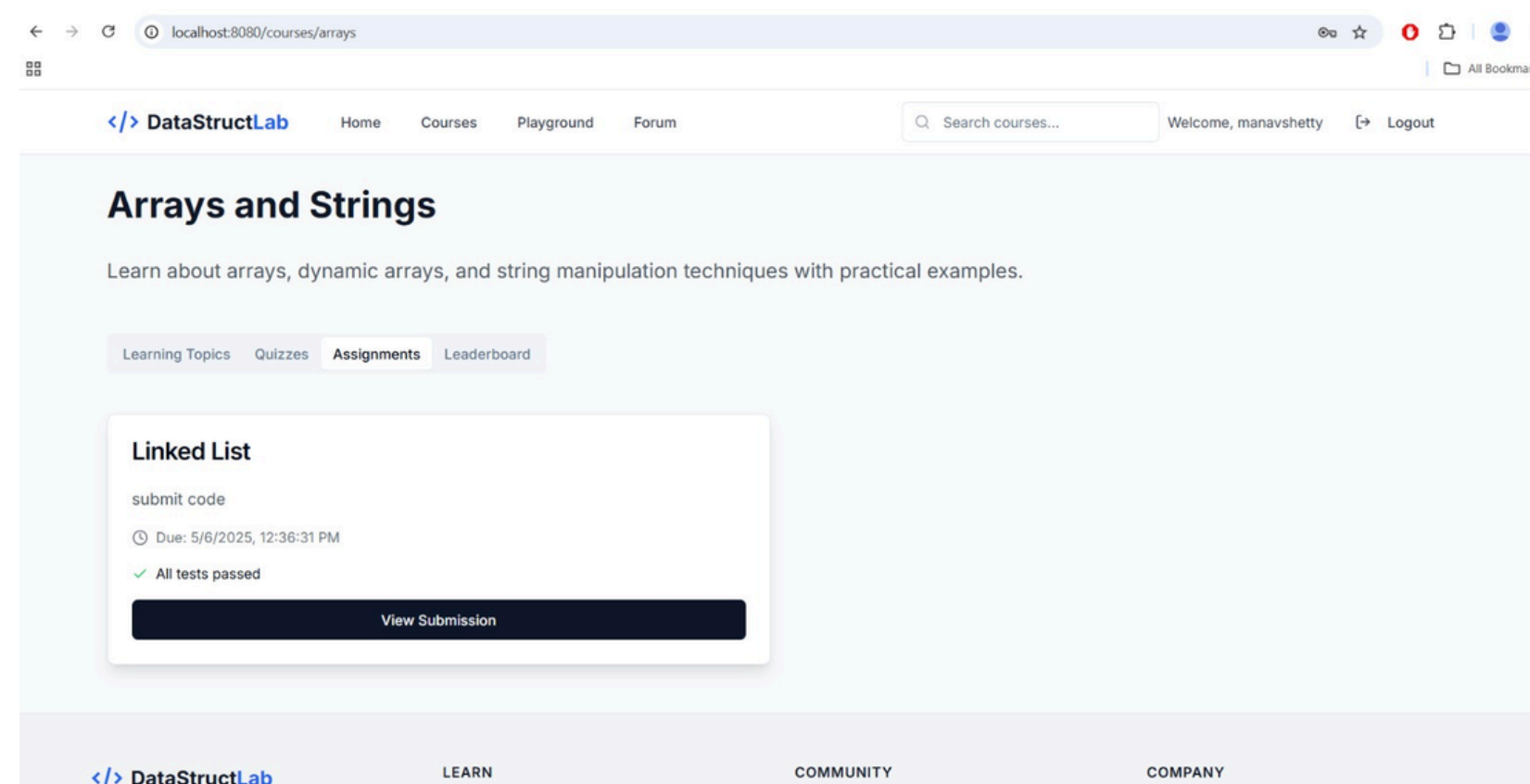
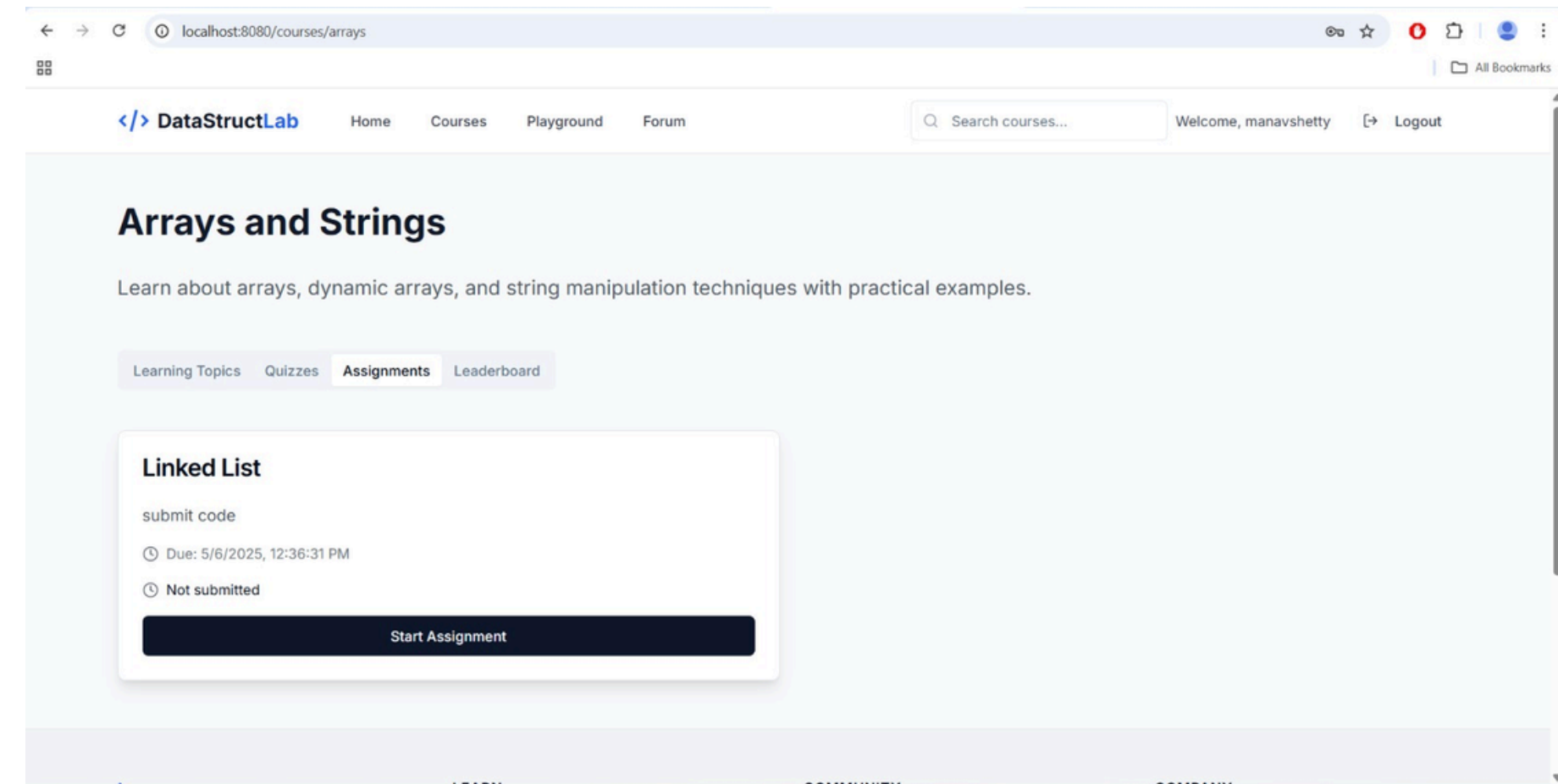
TEACHER LOGIN : LEADERBOARD

Leaderboard
showcasing
the score
along with
name of the
students.



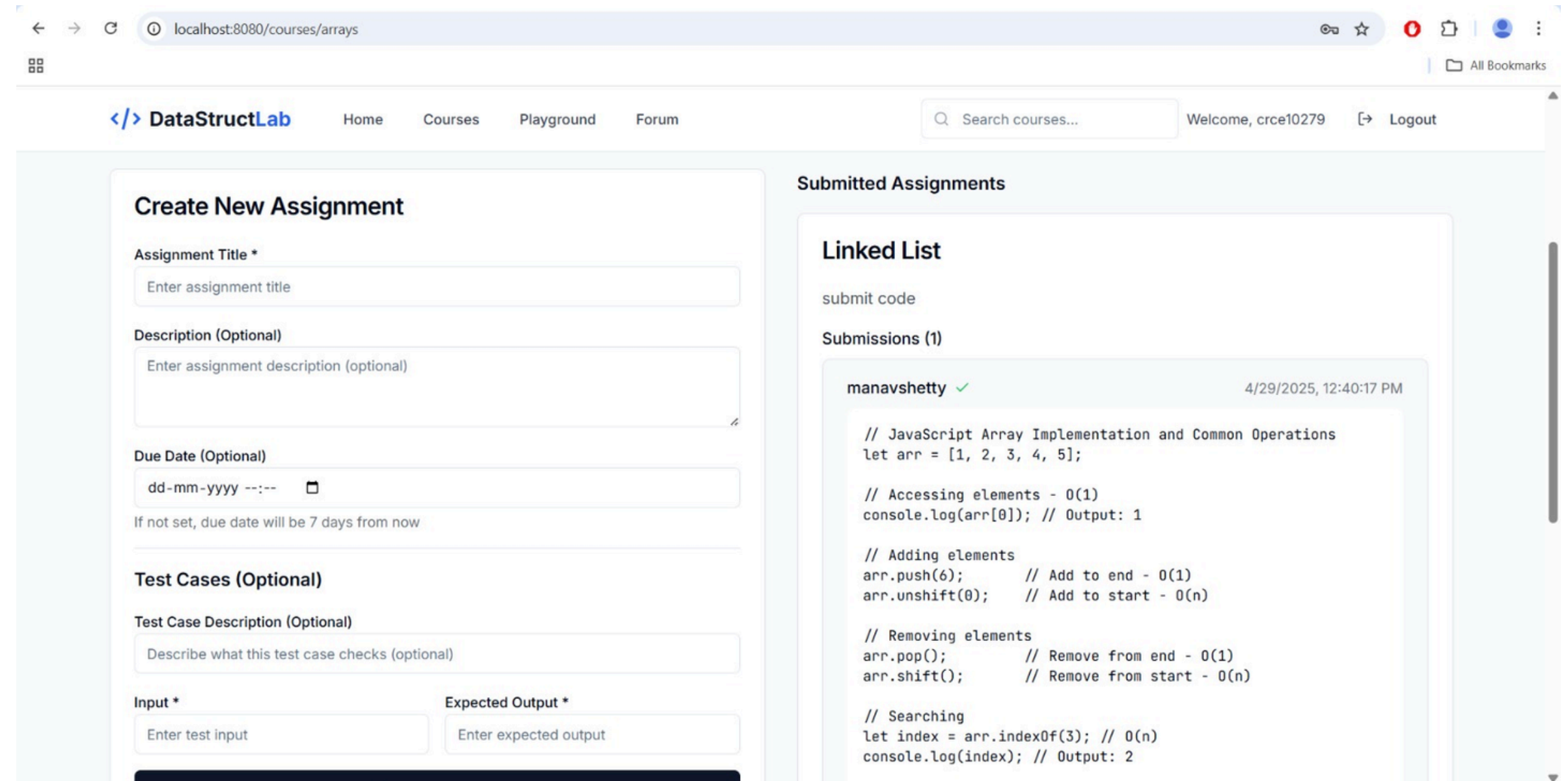
STUDENT LOGIN : ASSIGNMENTS

The
Assignment
created by
the teacher
are now
visible for
the students.



TEACHER LOGIN : ASSIGNMENTS

Once the
Assignment
is submitted
by the
student the
Teacher can
view the
submitted
assignment.



The screenshot displays the DataStructLab web application interface. The browser address bar shows 'localhost:8080/courses/arrays'. The navigation bar includes links for Home, Courses, Playground, and Forum, along with a search bar and user information (Welcome, crce10279) and a Logout button.

Create New Assignment

Assignment Title *
Enter assignment title

Description (Optional)
Enter assignment description (optional)

Due Date (Optional)
dd-mm-yyyy --:-- ☐
If not set, due date will be 7 days from now

Test Cases (Optional)
Test Case Description (Optional)
Describe what this test case checks (optional)

Input *
Enter test input

Expected Output *
Enter expected output

Submitted Assignments

Linked List

submit code

Submissions (1)

manavshetty ✓ 4/29/2025, 12:40:17 PM

```
// JavaScript Array Implementation and Common Operations
let arr = [1, 2, 3, 4, 5];

// Accessing elements - O(1)
console.log(arr[0]); // Output: 1

// Adding elements
arr.push(6); // Add to end - O(1)
arr.unshift(0); // Add to start - O(n)

// Removing elements
arr.pop(); // Remove from end - O(1)
arr.shift(); // Remove from start - O(n)

// Searching
let index = arr.indexOf(3); // O(n)
console.log(index); // Output: 2
```




THANK YOU

